

AttenGpKa: A Universal Predictor of Solvation Acidity Using Graph Neural Network and Molecular Topology

Hongle An, Xuyang Liu, Wensheng Cai,* and Xueguang Shao*



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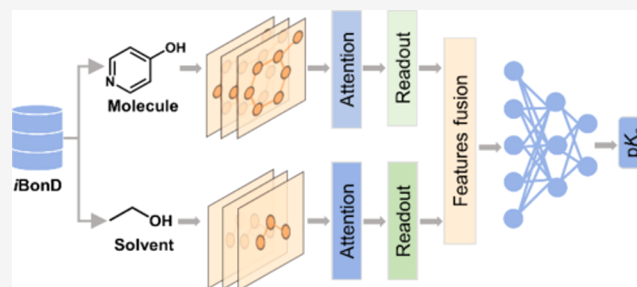


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ABSTRACT: Rapid and accurate calculation of acid dissociation constant (pK_a) is crucial for designing chemical synthesis routes, optimizing catalysts, and predicting chemical behavior. Despite recent progress in machine learning, predicting solvation acidity, especially in nonaqueous solvents, remains challenging due to limited experimental data. This challenge arises from treating experimental values in different solvents as distinct data domains and modeling them separately. In this work, we treat both the solutes and solvents equally from a perspective of molecular topology and propose a highly universal framework called AttenGpKa for predicting solvation acidity. AttenGpKa is trained using 26,522 experimental pK_a values from 60 pure and mixed solvents in the iBonD database. As a result, our model can simultaneously predict the pK_a values of a compound in various solvents, including pure water, pure nonaqueous, and mixed solvents. AttenGpKa achieves universality by using graph neural networks and attention mechanisms to learn complex effects within solute and solvent molecules. Furthermore, encodings of both solute and solvent molecules are adaptively fused to simulate the influence of the solvent on acid dissociation. AttenGpKa demonstrates robust generalization in extensive validations. The interpretability studies further indicate that our model has effectively learnt electronic and solvent effects. A free-to-use software is provided to facilitate the use of AttenGpKa for pK_a prediction.



support vector regression.¹⁷ Moreover, a set of continuous molecular descriptors, binary fingerprints, and fragment counts generated using PaDEL have been utilized in combination with three machine learning methods, support vector machines combined with k-nearest neighbors, XGBoost, and deep neural networks (DNN), to construct quantitative structure–activity relationships (QSAR) models for the accurate prediction of acid and base pK_a values in aqueous solutions.¹⁸

The pK_a value is highly sensitive to the solvent. However, most experimental data currently available are from aqueous solutions. Despite the scarcity of experimental data in non-aqueous solvents, researchers have made efforts to predict pK_a values in such solvents. Using descriptors that can describe the topological and chemical properties of molecules, Roszak et al. used graph convolutional neural networks (GCNN) for the first time to predict the pK_a value of C–H acid in DMSO.⁴ Yang et al. employed neural networks and XGBoost methods, and after careful screening of descriptors, they selected structural and

support vector regression.¹⁷ Moreover, a set of continuous molecular descriptors, binary fingerprints, and fragment counts generated using PaDEL have been utilized in combination with three machine learning methods, support vector machines combined with k-nearest neighbors, XGBoost, and deep neural networks (DNN), to construct quantitative structure–activity relationships (QSAR) models for the accurate prediction of acid and base pK_a values in aqueous solutions.¹⁸

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INTRODUCTION

Acid dissociation constant, pK_a , represents the ability of acidic or basic compounds in solution to lose or gain protons, and this value has profound implications for understanding the behavior of molecules in biological systems and their applications in fields such as chemical synthesis, materials science, and drug design.^{1–4} Under specific solvent conditions, the ionization state of small molecules directly affects their solubility, lipophilicity, stability, and ADMET (absorption, distribution, metabolism, excretion, toxicity) properties in biological systems.⁵ However, traditional experimental methods to determine pK_a values of analytes are complex and time-consuming, and high-level quantum mechanical (QM) calculations require substantial computational resources. Therefore, there is an urgent need for more efficient and accurate methods for predicting pK_a . In recent years, the application of machine learning methods in the field of chemistry has made significant advancements,^{6–15} providing a new paradigm for pK_a prediction.

A multitude of machine learning methods have been developed to predict the pK_a values of compounds in aqueous solutions.¹⁶ For predicting the pK_a values of fatty amines in aqueous solutions, methodologies have been established using root topological torsion fingerprints combined with five machine learning methods: random forests, extreme gradient boosting (XGBoost), partial least-squares, lasso regression, and

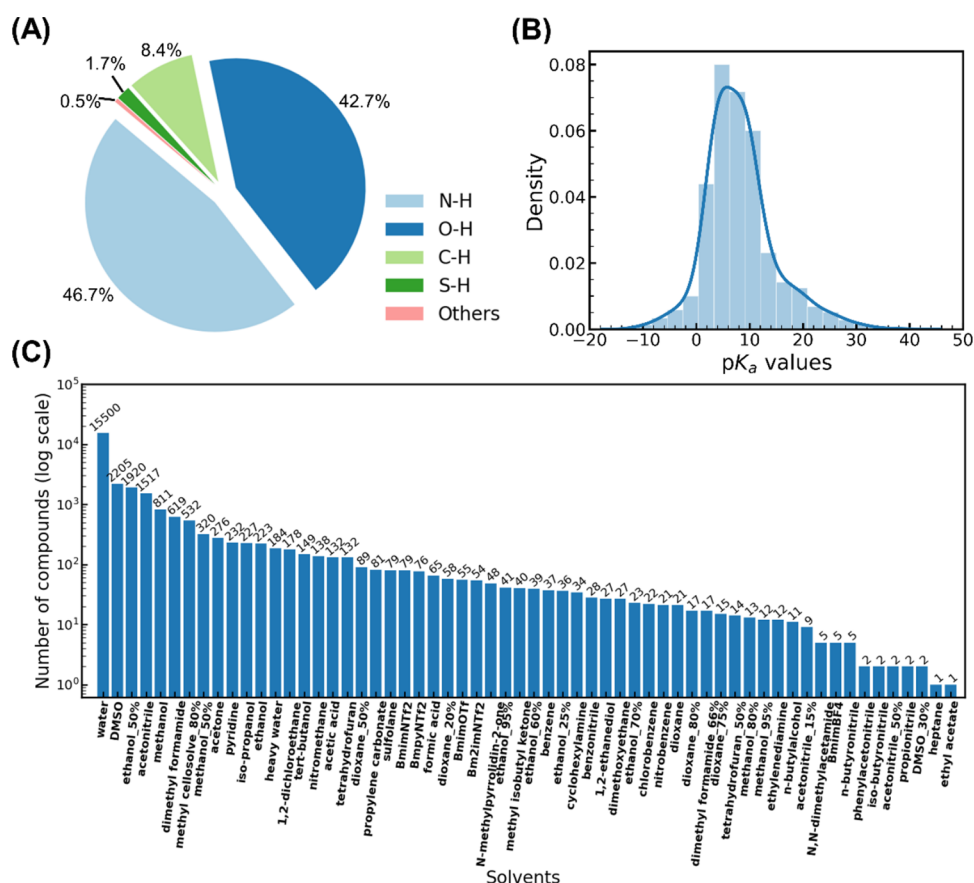


Figure 1. Characteristics of the data set. (A) Distribution of bond type at ionization site. (B) Distribution of pK_a values. (C) Counts of the compounds in different solvents.

physical organic parameters combined descriptors (SPOC) and ionic status labels (ISL) to establish a comprehensive pK_a prediction model.¹⁶ This model can predict the pK_a values of small molecules in five common solvents (water, DMSO, 50% ethanol, acetonitrile, and methanol). Both methods employ molecular descriptors to characterize molecules. However, selecting and defining descriptors is not only computationally time-consuming, but also likely to introduce human bias. In addition, the variety of solvents that can be predicted by these models is restricted.

With the rapid development of graph neural networks (GNN), the representation of molecules as molecular graphs is gaining increasing interest. In this approach, molecular structures are depicted as natural graphs, with atoms represented by nodes and bonds by edges.¹⁹ GNN can automatically extract molecular features associated with target properties through comprehensively learning information on atoms and bonds.^{20–26} In recent years, researchers have developed a series of GNN-based methods for predicting the pK_a of small molecules. MolGpka, built on GCNN, is used to predict the pK_a values of small molecules, and an associated web server for public use has also been developed.²⁷ However, due to the scarcity of experimental values, this method trains models using theoretically calculated data. The Graph- pK_a model, established based on GNN and combined with multiple instance learning, can accurately predict the macro- pK_a and micro- pK_a of compounds in aqueous solutions.²⁸ A limitation of this model is its inability to predict the pK_a values of C–H acids. MF-SuP- pK_a is a GNN model integrated with chemical knowledge, using strategies such as knowledge-guided data augmentation and

knowledge-aware subgraph pooling to achieve accurate prediction of pK_a in aqueous solutions.²⁹ In our previous work, we established the EEGpKa model based on message passing neural networks (MPNN).³⁰ This model introduces a knowledge-infused data augmentation technique, improving the prediction accuracy of pK_a for small sample data sets of C–H acids in both water and DMSO solvents.

Currently, there is still a lack of a general model that can accurately predict pK_a in water, pure nonaqueous, and mixed solvents simultaneously, without relying on molecular descriptors or quantum calculations. In this study, we introduced a novel and general framework, AttenGpKa, which combines the MPNN framework with attention mechanisms to accurately predict the pK_a values of compounds across various solvents. Based on the framework, the prediction model was established utilizing a large data set of pK_a experimental data from 60 solvents in the iBonD database.¹⁶ The framework treats both the target and solvent molecules as molecular graphs. These graphs are trained separately to learn the feature information, followed by attention mechanism based feature fusion to predict pK_a values. The robustness and accuracy of the model were verified through a series of internal and external tests. Through tests on phenolic compounds, it was confirmed that the model effectively learned the electronic effects of molecules and solvent effects. Key atoms or groups related to pK_a values were identified through interpretability study. The AttenGpKa model has been integrated into software, made available for free use, enabling accurate prediction of the pK_a values of compounds in different solvents.

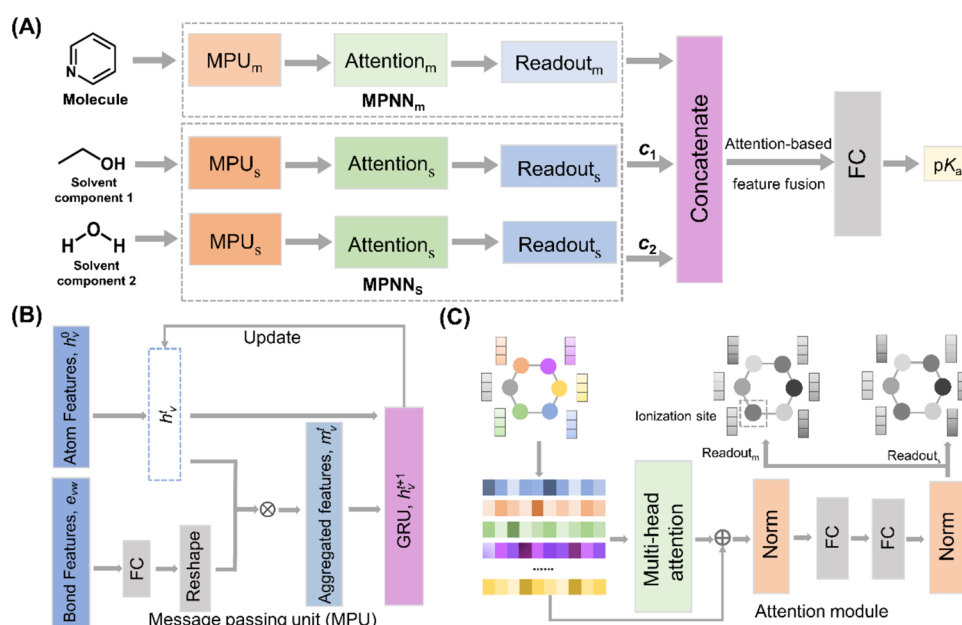


Figure 2. Overview of AttenGpKa. (A) Architecture of the MPNN-based prediction model. (B) Message passing unit (MPU). (C) Attention-based readout module.

DATA DESCRIPTION AND METHOD

Data Set. The iBonD 2.0 database, containing experimentally measured pK_a values of compounds in various solvents, was created by the iBonD team.¹⁶ It comprises over 20,000 compounds and more than 30,000 pK_a values. This database is freely available at <http://ibond.nankai.edu.cn>. The ionization site of each compound is annotated in the database. For polyprotic acids and bases, the ionization sites of each compound are labeled according to the order of acidity or basicity, with the pK_a values of each level of dissociation listed. Therefore, our model ultimately achieves micro-pK_a prediction. All data were manually processed and used as a data set for the AttenGpKa model. Since the data in the database originate from various sources, it will be the case that the same compound has different pK_a values in the same ionization state and solvent. In such cases, if the variance is greater than 2, the compound is eliminated. Otherwise, the average value is taken as the pK_a value of the compound. According to statistics, there are a total of 3328 molecules in the same ionization state and solvent with different pK_a values, among which 384 are excluded, and the remaining 2944 are averaged (more details can be found in the [Supporting Information](#)). Moreover, pK_a values below −12 or above 40 are discarded, as most of these are extrapolated from experimental values and lie outside the solvent pK_a window, leading to significant errors. Furthermore, data involving incorrect molecular structure drawings, damaged molecular images, and other such errors are also excluded from the data set. After data filtering, a data set comprising 26,522 pK_a values was obtained, which includes 19,721 molecules across different solvents. Specifically, there are 8460 protonated molecules, 1784 negatively charged molecules, and 9477 neutral molecules. The average Tanimoto similarity of molecular pairs of the entire data set was calculated using Morgan fingerprints³¹ with radii of 2, and the result was 0.09, which indicates the diversity of compound structures in the data set.

The distribution of the data set is illustrated in [Figure 1](#). [Figure 1A](#) presents the distribution of bond types at the ionization sites of compounds, with NH, OH, and CH ranking the top three in

terms of compound quantity. The distribution of pK_a values in the entire data set is depicted in [Figure 1B](#), while the distribution of pK_a values in different solvents is shown in [Figure S2](#). It can be observed from the figures that there is a wide range of pK_a values, indicating that the data set possesses a certain degree of representativeness. [Figure 1C](#) displays the number of compounds contained in 60 different solvents within the data set. Notably, water contains the highest number of pK_a values, reaching up to 15,500, while each nonwater solvent has relatively fewer pK_a values. The data set encompasses a wide variety of solvents, including protic solvents like water, methanol, isopropanol (iPrOH), and *tert*-butanol (tBuOH), as well as aprotic solvents such as DMSO, *N,N*-dimethylformamide (DMF), acetonitrile, nitromethane (NM), acetone, and *N,N*-dimethylacetamide (DMA), along with mixtures of different solvents. The pK_a values of the same compound vary significantly in different solvent environments. For instance, the pK_a value of 4-bromo-2-naphthoic acid is 8.85 in methanol, while it is 19.76 in acetonitrile. This variation is caused by the complex solvent environment, and the interactions between solvent and solute molecules during the dissociation process are challenging to explain with a clear and simple model.³²

Data Set Splitting. In this study, a 10-fold cross-validation method was adopted to evaluate the robustness of the model, ensuring the accuracy and generalization ability of the model for predicting pK_a values. The data set was divided using a random method. During the execution of the 10-fold cross-validation, the data set is evenly divided into ten parts, where each part takes turns as the test set, the fifth data subset is selected as the validation set for the adjustment of model parameters and performance optimization, and the remaining data is combined as the training set. The final ratio of the training, validation, and test sets is 8:1:1. To ensure the reproducibility of the model results reported in this study, all the data has been provided in the [Supporting Information](#).

Construction of AttenGpKa. Different MPNNs are employed by the AttenGpKa model to learn representations of both the target (subscript m) and solvent (subscript s)

Table 1. Results of 10-Fold Cross-Validation

metrics	1	2	3	4	5	6	7	8	9	10	average value ^a	standard deviation ^a
R	0.97	0.97	0.98	0.97	0.97	0.98	0.97	0.97	0.98	0.97	0.97	0.004
R ²	0.95	0.94	0.95	0.95	0.95	0.96	0.94	0.95	0.95	0.93	0.95	0.008
MAE	0.77	0.73	0.69	0.70	0.70	0.65	0.80	0.71	0.74	0.84	0.73	0.054
RMSE	1.45	1.48	1.43	1.46	1.43	1.17	1.69	1.49	1.40	1.64	1.46	0.131

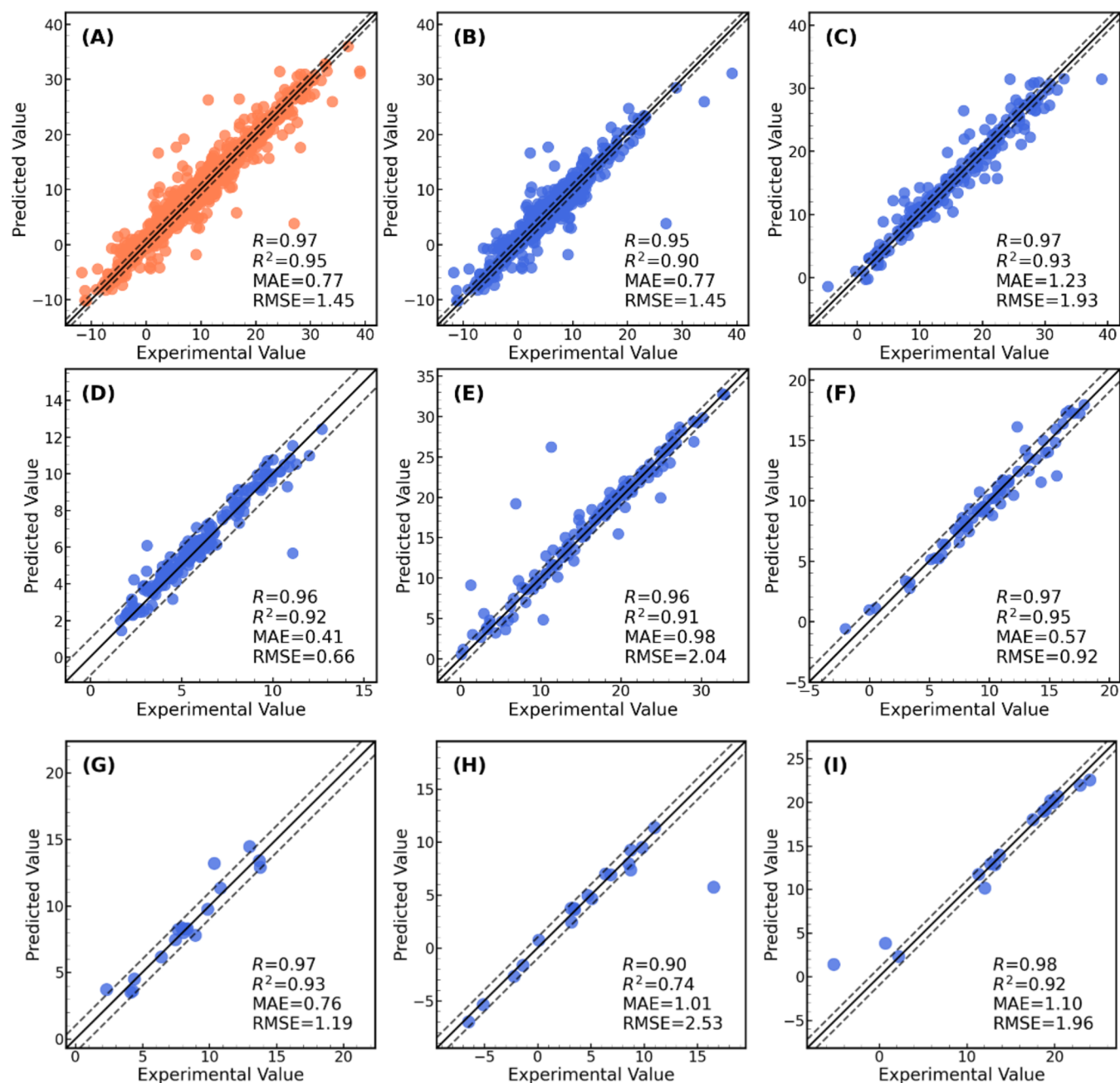
^aThe results of ten models in 10-fold cross-validation.

Figure 3. Correlation between the pK_a values predicted by AttenGpKa and obtained by experiment in different solvents. (A) All solvents. (B) Water. (C) DMSO. (D) 50% Ethanol. (E) Acetonitrile. (F) Methanol. (G) Ethanol. (H) Heavy water. (I) 1,2-Dichloroethane. Lines deviating by ± 1 pK_a unit are represented in black dashed lines.

molecules, as shown in Figure 2A.³³ Differing from classical approaches that treat various solvents as distinct data domains and model them separately,^{34,35} AttenGpKa incorporates the topology of solvent molecules into pK_a prediction, thereby

obviating the necessity to construct separate models for different solvents.

The pK_a prediction process of AttenGpKa is illustrated in Figure 2A, where molecules are described by atomic and bond vectors, which are encoded by the features from Table S1. The

Table 2. Performance of AttenGpKa and Other Methods on the Three External Test Sets^{a,b}

data set	method	model class	R ²	MAE	RMSE
SAMPL6	ACD/pK _a classic ^{41,42}	commercial	0.92	0.55	0.77
	MoKa ^{41,43}	commercial	0.88	0.77	0.90
	Epik Scan ^{41,44,45}	commercial	0.88	0.78	0.95
	Epik Microscopic ^{41,44,45}	commercial	0.87	0.81	1.03
	Jaguar ^{41,44,45}	commercial	0.88	0.80	1.10
	Chemicalize v18.23 ^{41,46}	commercial	0.87	1.04	1.27
	Hunt's model ⁴¹	academic	0.90	0.68	0.86
	Yang's NN ¹⁶	academic	0.82	0.85	1.17
	Yang's XGB ¹⁶	academic	0.84	0.80	1.07
	AttenGpKa	academic	0.87 (0.02)	0.74 (0.10)	0.94 (0.12)
SAMPL7	EC_RISM ³⁹	academic	0.93	0.53	0.72
	IEFPCM/MST ³⁹	academic	0.56	1.30	1.82
	DFT_M05-2X_SMD ³⁹	academic	0.03	2.28	2.90
	TZVP-QM ³⁹	academic	0.23	2.75	2.90
	Gaussian Process ³⁹	academic	0.30	2.91	3.49
	DFT_M06-2X_SMD ³⁹	academic	0.20	2.56	5.12
	Gaussian corrected ³⁹	academic	0.76	5.12	5.36
	Yang's NN ^{16,28}	academic	0.76	0.93	1.16
	Yang's XGB ^{16,28}	academic	0.52	1.48	1.62
	AttenGpKa	academic	0.82 (0.01)	1.00 (0.16)	1.24 (0.21)
Jensen et al.	PM3/COSMO ⁴¹	commercial	0.67	0.95	1.39
	Hunt's model ⁴¹	academic	0.84	0.60	0.98
	AttenGpKa	academic	0.78 (0.04)	0.77 (0.09)	1.17 (0.17)

^aThe results of Hunt's model for the SAMPL6 and Jensen et al. data sets, as well as the results of the PM3/COSMO model for the Jensen et al. data set, were calculated based on the data provided in the Supporting Information of the reference literatures. The remaining results were sourced from literatures. ^bValues in the parentheses represent the standard deviation of three repeated training runs.

learning of molecular topology is implemented by the message passing unit (MPU). As shown in Figure 2B, the atom h_v^0 is updated based on information from its neighboring atoms h_w^0 and bonds e_{vw} . The calculation is depicted in eqs 1 and 2, where W and b represent the weights and biases of the fully connected (FC) layer in the MPU, respectively, t denotes the time step of information updating, $N(v)$ represents a set composed of the v -th atom and its connected atoms, $R(\bullet)$ transforms the bond vector into a matrix with dimensions matching the atomic feature, and $\text{GRU}(\bullet)$ represents the gated recurrent unit,³⁶ which combines current-step features and aggregated features to update the atomic features.

$$m_v^t = \sum_{w \in N(v)} R(e_{vw}W + b)h_w^t \quad (1)$$

$$h_v^{t+1} = \text{GRU}(h_v^t, m_v^t) \quad (2)$$

After iteratively updating three times in the MPU, an attention-based readout module is employed to represent the molecule as a single vector. Equation 3 illustrates the calculation of the attention mechanism, where H is a matrix with rows representing the updated features of each atom, and W_Q , W_K , and W_V are the network weights. For solvent molecules, the readout vector is represented as the row average of H_R , as illustrated in eq 4. For target molecules, the readout vector is represented as a specific row of H_R , as illustrated in eq 5. As depicted in Figure 2C, utilizing the multihead attention mechanism,³⁷ atomic features are globally updated without considering molecular topology. The target molecule is characterized by the atomic features at its ionization site, while solvent molecules are represented by the average of all atomic features.

$$H_R = \text{soft max}[HW_Q(HW_K^T)](HW_V) \quad (3)$$

$$r_S = \frac{1}{N} \sum_{i=1}^N H_R^i \quad (4)$$

$$r_M = H_R^{i=K} \quad (5)$$

Subsequently, as shown in Figure 2A, solvent features are scaled by their molar percentage (c_1 and c_2) in the mixed solvent and combined with the target molecule features. A multihead attention mechanism is then utilized to integrate the features of the target and solvents molecules, and the final pK_a prediction is accomplished through two FC layers. The optimization of AttenGpKa is driven by minimizing the mean squared error (MSE) on the training data set, and L2 regularization is employed to mitigate overfitting. The training process of the model is shown in Figure S3. All network hyperparameters optimized are listed in Table S2 in the Supporting Information.

Evaluation. To verify the accuracy of the model, a series of internal, external, and instance tests were conducted. Internal validation was achieved through 10-fold cross-validation, while external validation employed three external benchmark data sets, namely SAMPL6,³⁸ SAMPL7,³⁹ and a data set published by Jensen et al.⁴⁰ The performance of these data sets was comprehensively evaluated using four metrics: Pearson correlation coefficient (R), coefficient of determination (R^2), mean absolute error (MAE), and root mean squared error (RMSE). The higher the R and R^2 values, the lower the MAE and RMSE values, the better the prediction performance of the model. Furthermore, the accuracy of the AttenGpKa model was also verified by predicting the pK_a values of phenol compounds with similar structures in various solvents.

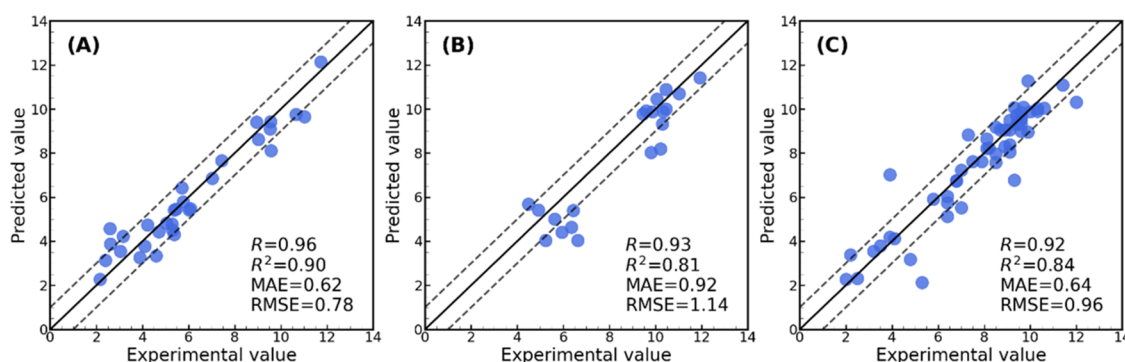


Figure 4. Correlation between the pK_a values predicted by AttenGpKa and obtained by experiment on the three external test sets. (A) SAMPL6. (B) SAMPL7. (C) Jensen et al. data set. Lines deviating by ± 1 pK_a unit are represented in black dashed lines.

Table 3. Experimental and Predicted Values of Some Phenolic Compounds in Different Solvents

compounds	water	DMSO	acetonitrile	methanol
3,5-dichlorophenol	7.95/8.18 ^a	13.64/13.01	23.17/23.31	12.01/12.94
3,5-dinitrophenol	6.57/6.66	10.50/10.60	20.24/20.50	10.17/10.20
3-nitrophenol	8.59/8.36	13.93/13.75	23.37/23.85	12.52/12.40
3-trifluoromethyl-4-nitrophenol	6.05/6.41	8.84/9.30	19.26/19.30	— ^b
3-trifluoromethylphenol	9.15/9.04	15.27/14.30	25.48/24.90	13.32/12.10
compounds	DMF	<i>i</i> PrOH	<i>t</i> BuOH	
3,5-dichlorophenol	—	12.25/14.05	16.88/17.04	
3,5-dinitrophenol	11.08/11.30	9.90/10.84	13.21/13.40	
3-nitrophenol	14.07/13.85	12.53/13.92	16.88/16.99	
3-trifluoromethyl-4-nitrophenol	9.74/10.40	9.29/9.90	12.86/12.77	
3-trifluoromethylphenol	15.48/15.70	13.42/12.50	18.06/17.10	

^aThe left side of the “/” represents the predicted value, and the right side represents the experimental value. ^b— indicates that no experimental value is found.

RESULTS AND DISCUSSION

Model Performance. To assess the robustness of the model developed for predicting the pK_a values of compounds, a 10-fold cross-validation method was employed to systematically analyze the entire data set. The evaluation metrics for all test sets are summarized in Table 1. It was observed that each test set exhibited high R and R^2 values, while MAE and RMSE were comparatively lower, and that these metrics varied slightly between different test sets. The average and standard deviation over the results of all test sets were observed to be small, further confirming the high consistency and accuracy of the model across different data subsets. Furthermore, the model for the third fold was trained seven times to evaluate the uncertainty. Specifically, the standard deviation of RMSE on test data set is 0.09. The results indicate that the model possesses high accuracy and robustness, is not easily affected by specific data division, and has strong generalization capability.

The first test set of cross-validation is taken as an example to demonstrate the prediction performance of the AttenGpKa model in various solvents. Figure 3 shows the correlation results between predicted and experimental values of compounds in this test set in different solvents. Figure 3A represents the results for the entire test set, while Figure 3B–I shows the results in eight different solvents. The results in other solvents, presented in Figure S4 in the Supporting Information. The results indicate that the model demonstrates good predictive capability, both for the overall test set and for each specific solvent. Even for solvents with fewer training samples, the AttenGpKa model can accurately predict the pK_a values of compounds, which proves its effectiveness in extracting features related to pK_a values and

its generalizability. Furthermore, the reproducibility of the model for the nonlinear relationship between the mixture composition and pK_a value is examined. As seen in Figure S5 in the Supporting Information, the results indicate that this nonlinear relationship can be effectively learned by the model.

Comparison of Benchmarking Data Sets. To further validate the accuracy of the model, three external benchmark data sets were utilized for evaluation: SAMPL6,³⁸ SAMPL7,³⁹ and a data set published by Jensen et al.⁴⁰ The first two data sets are part of the pK_a prediction challenges organized by the Drug Design Data Resource Community. SAMPL6 comprises 24 drug-like small molecules (17 drug-like fragments and 7 drug-like molecules), encompassing a total of 31 pK_a experimental values. Compared to SAMPL6, SAMPL7 presents a more diverse and complex data set with 22 varied compounds, primarily sulfonamides, though two lack experimental pK_a values. The data set published by Jensen et al. includes 53 pK_a values for 48 amine-containing pharmaceutical molecules.

The chemical structures of the compounds in the three data sets, along with their experimental and predicted pK_a values, are presented in Figures S6–S8 in the Supporting Information. The performance of the AttenGpKa model compared to other models is shown in Table 2 and Figure 4. It can be observed that on the SAMPL6 data set, the performance of AttenGpKa is comparable to that of the other methods listed in the table. Compared to the XGB method proposed by Yang et al.,¹⁶ AttenGpKa shows a performance improvement of nearly 12% as quantified by RMSE, and a 20% improvement compared to the NN model. Furthermore, for compounds with large predictive deviations from the model proposed by Yang et al., such as

SM03 ($\Delta pK_a = 2.0$), SM05 ($\Delta pK_a = 1.99$), SM14 ($\Delta pK_a = 2.45$), SM18a ($\Delta pK_a = -3.12$), and SM24 ($\Delta pK_a = 1.58$), AttenGpKa predicts their ΔpK_a values of -0.15 , -1.25 , 1.99 , -1.46 , and 1.28 , respectively. It can be seen that significant progress has been made, but there is still room for further improvement, which may be due to the lack of this type of compounds in the data set. Compared to other models submitted for SAMPL7, the performance of AttenGpKa ranks second, with the highest being EC_RISM (embedded cluster reference interaction site model), a QM-based method. AttenGpKa performs slightly better than Yang's XGB model and is comparable in performance to Yang's NN model. Using the entire data set published by Jensen et al., without excluding the cefadroxil molecule, our model exceeds the PM3/COSMO method.⁴¹

Predicting pK_a of Phenols in Different Solvents. To further validate the accuracy of our model in different solvents, a selection of phenolic compounds was chosen for prediction.³² Table 3 presents the predicted and experimental values in various solvents. It can be seen that even in solvents with fewer compounds in the data set, such as *i*PrOH and *t*BuOH, good predictive performance was achieved. This demonstrates that our model has sufficiently learned the characteristic information on molecules, enabling it to accurately predict pK_a values in different solvents.

Electronic Effect Learned by AttenGpKa. The electronic effects of compounds are closely related to their pK_a values. To verify that the AttenGpKa model has learned the electronic effects, a set of C–H acids in the data set in aqueous solutions were selected for validation, with the requirement that the carbon atom must be connected to at least two hydrogen atoms. Ultimately, 1131 compounds were screened out. One of the hydrogen atoms on the carbon atom was replaced with groups that have significant inductive (I) and conjugative (C) effects to observe changes in the predicted pK_a values. Groups with -I and -C effects can increase acidity, lowering the pK_a value; conversely, groups with +I and +C effects decrease acidity, raising the pK_a value. If the trend of pK_a value changes compared to the original compounds is consistent, it can be qualitatively stated that the model has learned the electronic effects. Accuracy can be obtained by calculating the percentage of compounds exhibiting consistent trend changes out of the total number of compounds. As shown in Figure 5, the accuracy of groups with -I effects such as $-\text{NO}_2$, $-\text{SO}_2\text{CH}_3$, $-\text{CN}$, $-\text{SO}_3\text{H}$, $-\text{COOH}$, and $-\text{COCl}$ is over 90%, and that of groups with -I effects, including $-\text{CHO}$, $-\text{CONH}_2$, and $-\text{CF}_3$, is also more than 80%. The accuracy of groups with -C effects like $-\text{C}_6\text{H}_4\text{NO}_2$, $-\text{C}_6\text{H}_4\text{CN}$, $-\text{CH}=\text{CH}-\text{CH}=\text{CHF}$, and $-\text{CH}=\text{CH}-\text{CH}=\text{CHCl}$ is above 70%. Groups with +I effects, including $-\text{C}(\text{CH}_3)_3$, $-\text{CH}(\text{CH}_3)_2$, $-\text{CH}_2\text{CH}_3$, and those with +C effects like $-\text{C}_6\text{H}_4\text{NH}_2$, $-\text{C}_6\text{H}_4\text{OCH}_3$, have an accuracy of over 60%. These results suggest that the AttenGpKa model has learned the electronic effects to a certain degree.

To further validate that the model has learned the electronic effects, the AttenGpKa model was utilized to predict the pK_a values of phenol and a series of its substituted derivatives in DMSO solvent, as illustrated in Figure 6. Figure 6A displays the predicted and experimental pK_a values for phenol substituted at the para position with groups having different electronic effects. Compared to phenol, the substitution by a group with a -I effect, such as $-\text{NO}_2$, $-\text{CN}$, $-\text{Cl}$, $-\text{Br}$, is observed to lower the pK_a values, whereas the substitution by a group with a +I effect, $-\text{CH}_3$, is found to raise the pK_a values. The change in the pK_a

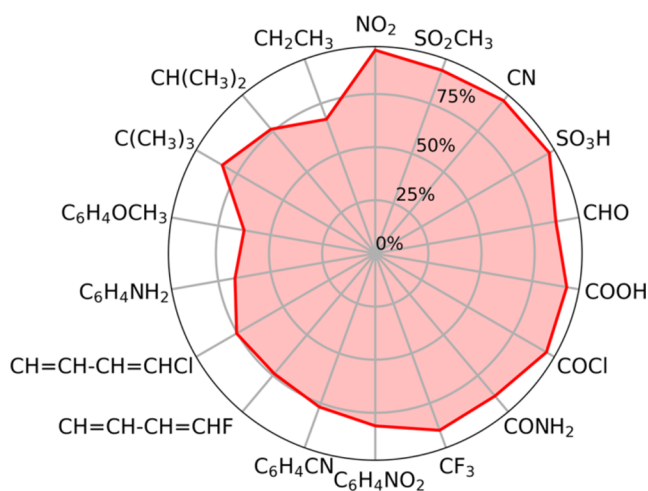


Figure 5. Accuracy of AttenGpKa predictions for trends following substitution with groups having different electronic effects.

value of the substituted compound relative to its original one is defined as $\Delta pK_a^s - ^\circ$. It is observed that the prediction values for these compounds align with the experimental trends, and the $\Delta pK_a^s - ^\circ$ values are also close, indicating that the model can accurately learn the influence of electronic effects on pK_a . Figure 6B shows the experimental and predicted values for phenol substituted with different numbers of $-\text{NO}_2$ groups. The experimental pK_a values decrease with an increasing number of $-\text{NO}_2$ groups, due to the strong -I effect of this group, leading to increased acidity of the compound.⁴⁷ The predicted values are found to be very close to the experimental values, and the $\Delta pK_a^s - ^\circ$ values are also similar. This indicates that our model has effectively learned electronic effects. The predicted results for these compounds in water solvent, shown in Figure S9, are essentially consistent with those in DMSO solvent.

Interpretability of AttenGpKa. The black-box nature of deep learning has always been a challenge.^{48,49} To address this issue, many interpretable methods have been developed, among which perturbation-based methods are commonly used.^{50,51} By perturbing different input features and monitoring the changes in predictions, key features can be identified. To enhance the interpretability of the AttenGpKa model, each heavy atom is masked to observe the corresponding changes in pK_a values, thereby identifying key atoms or groups related to pK_a values. Figure 7 displays the visualization results of the contributions of different atoms in some representative molecules, with other representative molecular visual structures seen in the Supporting Information Figure S10. Figure 7A shows charged molecules, revealing that the atoms bearing charges have the greatest impact on the pK_a values of different compounds, while other atoms have relatively smaller impacts. Figure 7B–E displays compounds with $-\text{NO}_2$, $-\text{SO}_2\text{R}$, $-\text{CN}$, $-\text{COOH}$ and $-\text{COR}$ groups, respectively, showing significant contributions of these groups to pK_a values, even though these groups are not always adjacent to the atoms of the ionization sites in some compounds. These electron-withdrawing groups affect the distribution of the electron cloud in the compound,^{47,52–58} which in turn affects the ability of the compound to lose or gain protons. On the other hand, it can be seen that the AttenGpKa model is capable of capturing the contributions of atoms far from the ionization sites, indicating that the model can effectively learn the global information on small molecules. In addition, the attention scores of each atom after Readout_m process were projected onto the

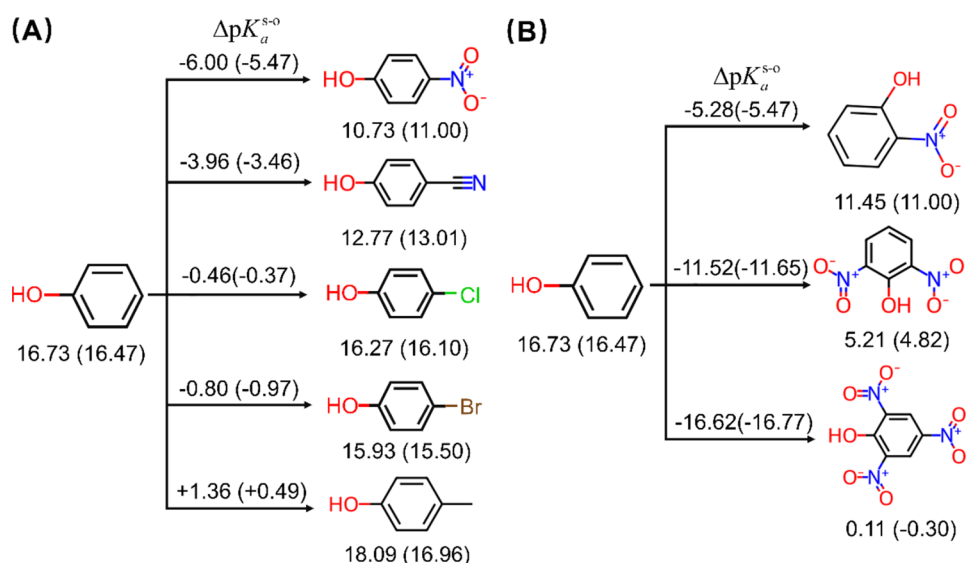


Figure 6. Experimental and predicted pK_a for phenol and substituted phenol in DMSO solvent. (A) Phenol with various substituent groups. (B) Nitro-substituted phenol. Experimental values are enclosed in brackets, and ΔpK_a values are displayed above the arrows.

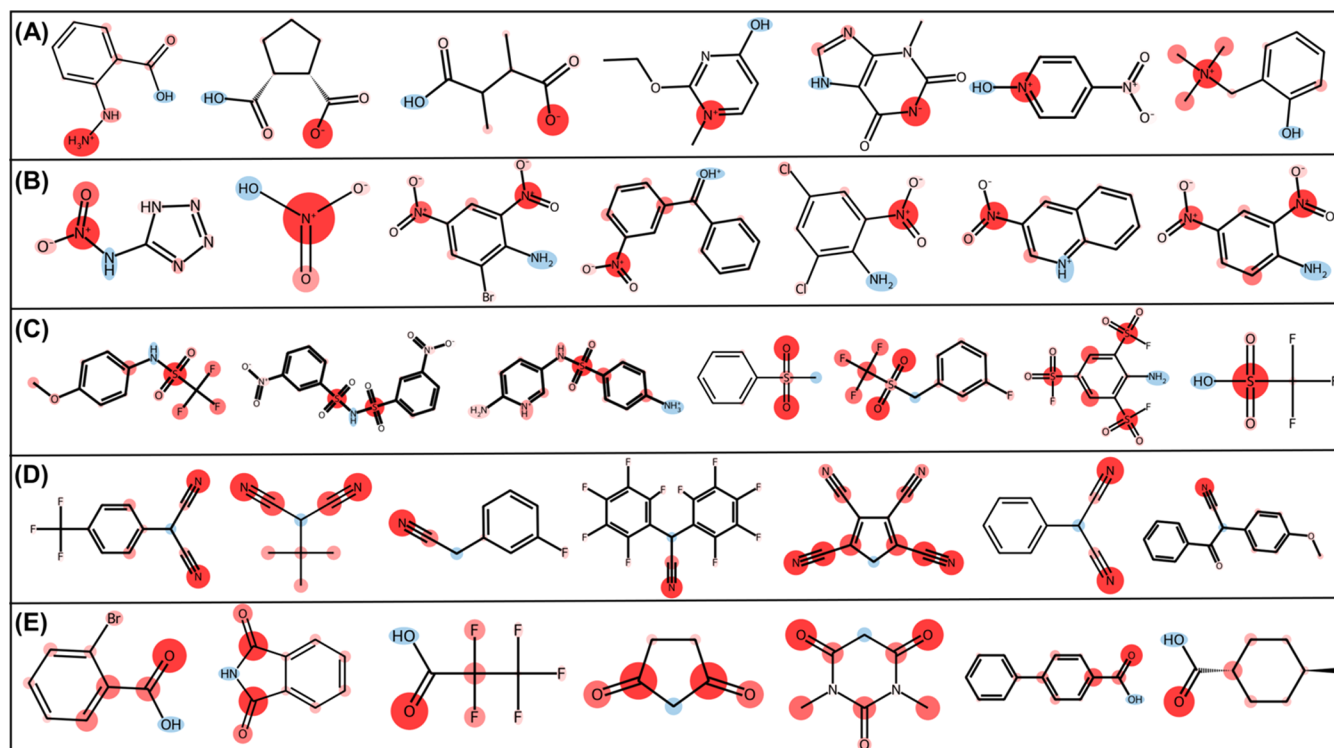


Figure 7. Visual results depicting atomic contributions to pK_a values in some representative molecules using atomic masking technology. Atoms highlighted in blue represent the ionization sites, while the red color indicates the level of contribution for each atom. A darker and larger red corresponds to a greater contribution. (A) Charged molecules. (B) $-\text{NO}_2$. (C) $-\text{SO}_2\text{R}$. (D) $-\text{CN}$. (E) $-\text{COR}$ or $-\text{COOH}$.

corresponding molecules, as shown in Figure 8. The contribution of each atom to the pK_a value can be observed, and the results are consistent with the above.

Applicability Domain Analysis. The applicability domain (AD), which refers to the scope within which a model can be applied, is influenced by the categories of compounds and structural characteristics encompassed within the training set. The AD of the AttenGpKa model is defined based on the similarity of molecules in the training set, as determined by Euclidean distance using Morgan fingerprints.³¹ Specific details

can be found in our previous work.³⁰ By calculating the distance matrix of compounds in the training set, the average Euclidean distance $\bar{d}$ and the standard deviation σ of these distances are obtained, which are 6.44 and 2.95, respectively. The distance threshold D_T for describing the AD is calculated using $D_T = \bar{d} + Z \cdot \sigma$, where Z is an empirical parameter representing the significance level. Combined with the number of compounds outside the AD (OAD) (Table 4) and the performance of the model in the inside the AD (IAD) and OAD (Table 5) under

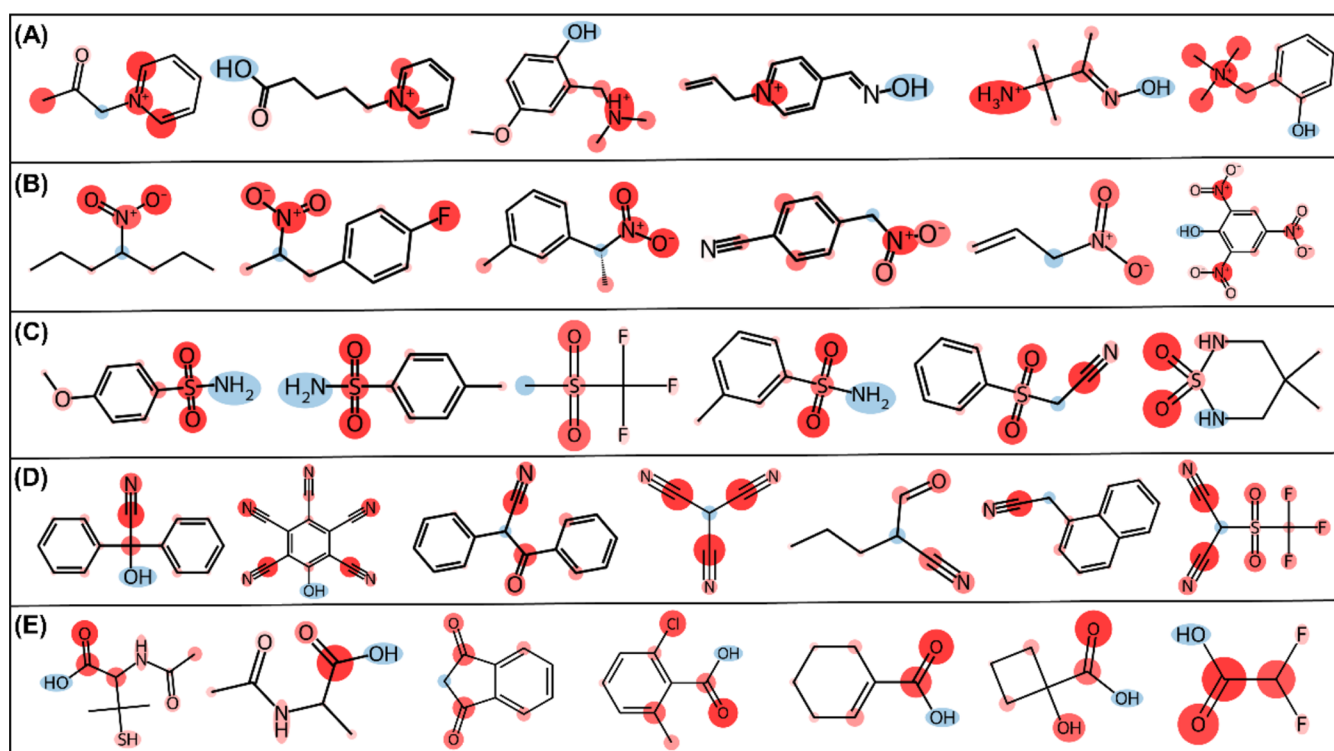


Figure 8. Visual results depicting atomic contributions to pK_a values in some representative molecules by attention mechanism. Atoms highlighted in blue represent the ionization sites, while the red color indicates the level of contribution for each atom. A darker and larger red corresponds to a greater contribution. (A) Charged molecules. (B) $-\text{NO}_2$. (C) $-\text{SO}_2\text{R}$. (D) $-\text{CN}$. (E) $-\text{COR}$ or $-\text{COOH}$.

Table 4. Number of Outside the AD (OAD) Compounds in the Test Set with Different Parameters

		Z				
		1.0	1.2	1.4	1.6	1.8
k	1	325	256	210	176	144
	3	396	309	246	211	185
	5	449	355	292	234	206
	7	488	386	309	261	216
D_T		8.97	9.49	10.01	10.52	11.04

different parameters, it is ultimately determined that when $Z = 1.6$ and $k = 3$, $D_T = 10.52$ is the AD for the AttenGpKa model.

CONCLUSIONS

In the present contribution, based on 26,522 pK_a values in 60 solvents from the *iBonD* database, a general model, AttenGpKa, was established for the accurate prediction of solvation acidity of compounds in different solvents. The target and solvent molecules were treated as molecular graphs, which avoided the complex manual engineering of selecting and defining molecular descriptors. By integrating the advantages of GNN and attention mechanisms, AttenGpKa achieves accurate predictions of pK_a in pure water, pure nonaqueous, and mixed solvents. Through a series of internal and external tests, the accuracy and robustness of the model were proven. In addition, tests on phenolic compounds demonstrated that the model can learn solvent effects and accurately predict pK_a values in various solvents. Furthermore, it is proved that this model can effectively learn the electronic effects of compounds from both qualitative and quantitative perspectives. By masking each atom in the compounds and observing changes in the predicted pK_a values, key atoms or groups closely related to pK_a values are identified,

Table 5. Predictive Performance of Compounds Inside and Outside the AD in the Test Set under Different Parameters

parameters (Z, k)	R^2		MAE		RMSE	
	IAD	OAD	IAD	OAD	IAD	OAD
(1.0, 1)	0.96	0.96	0.66	0.89	1.33	1.43
(1.0, 3)	0.96	0.96	0.66	0.85	1.34	1.36
(1.0, 5)	0.95	0.97	0.66	0.82	1.35	1.31
(1.0, 7)	0.95	0.96	0.66	0.82	1.34	1.35
(1.2, 1)	0.96	0.96	0.66	0.96	1.32	1.52
(1.2, 3)	0.96	0.96	0.66	0.89	1.33	1.42
(1.2, 5)	0.96	0.96	0.66	0.87	1.34	1.38
(1.2, 7)	0.96	0.96	0.66	0.85	1.34	1.35
(1.4, 1)	0.96	0.96	0.66	0.98	1.33	1.54
(1.4, 3)	0.96	0.96	0.66	0.93	1.33	1.47
(1.4, 5)	0.95	0.97	0.67	0.87	1.34	1.38
(1.4, 7)	0.96	0.96	0.66	0.89	1.34	1.41
(1.6, 1)	0.96	0.96	0.67	0.94	1.33	1.49
(1.6, 3)	0.96	0.96	0.66	0.98	1.33	1.54
(1.6, 5)	0.96	0.96	0.66	0.95	1.33	1.50
(1.6, 7)	0.96	0.96	0.67	0.90	1.33	1.44
(1.8, 1)	0.96	0.96	0.68	0.92	1.34	1.50
(1.8, 3)	0.96	0.96	0.67	0.91	1.34	1.45
(1.8, 5)	0.96	0.96	0.67	0.92	1.33	1.46
(1.8, 7)	0.96	0.96	0.67	0.90	1.34	1.44

which not only helps to explain the internal mechanism of the neural network as a black box model, but also provides chemical insights for understanding the pK_a value of the compound. We offer a user-friendly software to facilitate the use of AttenGpKa, and also provide the source code.

■ ASSOCIATED CONTENT

Data Availability Statement

All data are described in the [Data Description and Method](#) section of the article. The building, training and testing of the model were implemented using TensorFlow 2.11.0, Keras 2.11.0 and Python 3.8, which can be obtained free of charge from <https://pypi.org/project/tensorflow/2.11.0/>, <https://pypi.org/project/keras/2.11.0/>, and <https://www.python.org/downloads/release/python-380/>, respectively. RDKit (<https://rdkit.org/docs/index.html>) was used for processing all molecules. All the data has been provided in the Supporting Information. All the code and software files have been deposited at Zenodo (<https://zenodo.org/records/11436576>).

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.4c00449>.

Descriptions of atomic and bond features (Table S1); optimal hyperparameters of AttenGpKa (Table S2); operational steps of the AttenGpKa software for predicting the pK_a values of molecules in different solvents (Figure S1); distribution of pK_a values in different solvents (Figure S2); training process of AttenGpKa (Figure S3); predicted pK_a values of compounds in ethanol–water solutions at various mole percentages (Figure S4); correlation between the pK_a values predicted by AttenGpKa and obtained by experiment in different solvents (Figure S5); predicted pK_a values of the compounds in SAMPL6, SAMPL7, and Jensen et al. data sets using the AttenGpKa model (Figures S6–8); experimental and predicted pK_a for phenol and substituted phenol in water solvent (Figure S9); visual results depicting atomic contributions to pK_a values in other representative molecules (Figure S10) (PDF)

Data files for the AttenGpKa model in this study (ZIP)

■ AUTHOR INFORMATION

Corresponding Authors

Wensheng Cai – Research Center for Analytical Sciences, Tianjin Key Laboratory of Biosensing and Molecular Recognition, State Key Laboratory of Medicinal Chemical Biology, College of Chemistry, Nankai University, Tianjin 300071, China; Haihe Laboratory of Sustainable Chemical Transformations, Tianjin 300192, China; orcid.org/0000-0002-6457-7058; Email: wscai@nankai.edu.cn

Xueguang Shao – Research Center for Analytical Sciences, Tianjin Key Laboratory of Biosensing and Molecular Recognition, State Key Laboratory of Medicinal Chemical Biology, College of Chemistry, Nankai University, Tianjin 300071, China; Haihe Laboratory of Sustainable Chemical Transformations, Tianjin 300192, China; orcid.org/0000-0001-5027-4382; Email: xshao@nankai.edu.cn

Authors

Hongle An – Research Center for Analytical Sciences, Tianjin Key Laboratory of Biosensing and Molecular Recognition, State Key Laboratory of Medicinal Chemical Biology, College of Chemistry, Nankai University, Tianjin 300071, China; Haihe Laboratory of Sustainable Chemical Transformations, Tianjin 300192, China; orcid.org/0000-0003-2243-2827

Xuyang Liu – Research Center for Analytical Sciences, Tianjin Key Laboratory of Biosensing and Molecular Recognition, State

Key Laboratory of Medicinal Chemical Biology, College of Chemistry, Nankai University, Tianjin 300071, China; Haihe Laboratory of Sustainable Chemical Transformations, Tianjin 300192, China; orcid.org/0000-0003-1576-0136

Complete contact information is available at:

<https://pubs.acs.org/doi/10.1021/acs.jcim.4c00449>

Notes

The authors declare no competing financial interest.

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